16) If $A = p^2 + 3p + 5$ and $B = 2p^2 - 5p - 7$ then find:

(a) $2A + 3B$ (b) $A - B$

17) Add: $l^2 - m^2$, $m^2 - n^2$ and $n^2 - l^2$

18) Simplify: $8(2a + 3b - c) - 10(a + 2b + 3c)$

19) Simplify: $(7x + 3y) - (2x - 13y) + (6x - 4y)$

20) What should be added to $4x^2 - 5xy - 9$ to get $-3x^2 + 4xy + 5$?

21) Simplify $3x(4x - 5) + 3$ and find the values for: (i) $x = 1$ (ii) $x = \frac{1}{2}$

22) Simplify the expression and find the value if $x = -2$ and $y = 1$ $4(x + y) + 7y - 4x - 3$

23) If $a = 5$ and $b = -3$, evaluate $4a^2 + 5b - 3$.

24) Fill in the blank:

The value of the expression $2a - 3b + abc$ if $a = 0, b = -2, c = -1$ is _____.

25) Find the value of c in $2a + 3b - c = 12$, if $a = 2$ and $b = 3$.

26) Find the value of the expression $3xy - 4yz$ if $x = 2, y = -1$ and $z = 3$.

27) Evaluate $4a + 3b - 2c$ when $a = 4$; $b = 3$; $c = 5$. 16 + 9 + 10

28) If $p = -2$, find the value of $-2p^3 - 3p^2 + 4p + 7$.

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SIMPLE EQUATIONS

1. Write the following statements in the form of equations:
(i) The sum of three times x and 11 is 32. $= 3x+11=32$
(ii) If you subtract 5 from 6 times a number, you get 7. $6x-5=7$
(iii) One fourth of m is 3 more than 7. $\frac{1}{4}m=3+7$
(iv) One third of a number plus 5 is 8. $\frac{1}{3}n+5=8$

2. Convert the following equations in statement form:

(i) $x-5=9$ (ii) $5p=20$ (iii) $3n+7=1$ (iv) $\frac{m}{5}-2=6$.

5 taken away from x is 9

3. Write the following situation in the form of equations:
Raju's father's age is 5 years more than three times Raju's age. Raju's father is 44 years old. Set up an equation to find Raju's age. $(f-5)=3$ $\frac{f-5}{3}$

4. A shopkeeper sells mangoes in two types of boxes, one small and one large. A large box contains as many as 8 small boxes plus 4 loose mangoes. Set up an equation which gives the number of mangoes in each small box. The number of mangoes in a large box is given to be 100. $\frac{(100-4)}{8}$

5. Write equations for the following statements:
(i) The sum of numbers x and 4 is 9. $x+4=9$
(ii) The difference between y and 2 is 8. $y-2=8$
(iii) Ten times a is 70. $10a=70$
(iv) The number b divided by 5 gives 6. $\frac{b}{5}=6$
(v) Three fourth of t is 15. $t\frac{3}{4}=15$
(vi) Seven times m plus 7 gets you 77. $7m=77$
(vii) One fourth of a number minus 4 gives 4. $n\frac{1}{4}-4=4$
(viii) If you take away 6 from 6 times y, you get 60. $7=6y-6=8$
(ix) If you add 3 to one third of z, you get 30. $z\frac{1}{3}=30$

6. Write the following statements in the form of equations:
(a) 11 added to $2m$ to get 40. $= 11+2m=40$
(b) 11 subtracted from $2m$ to 25 $= 2m-11=25$
(c) 5 times y to which 3 is added to get 45 $5y+3=45$
(d) 5 times y from which 3 is subtracted to get 33 $5y-3=33$
(e) y is multiplied by – 8 to get 24 $-8y=24$
(f) y is multiplied by – 8 and then 5 is added to the result to get 29. $-8y+5=29$
(g) y is multiplied by 5 and the result is subtracted from 16 to get 4 $16-5y=4$
(h) y is multiplied by – 5 and the result is added to 16 to get 8. $-5y+16=8$

7. The length of a rectangular hall is 4 meters less than 3 times the breadth of the hall. What is the length, if the breadth is b meters? $L=4-3b$

8. Solve: (a) $3n+7=25$ $n=6$ (b) $2p-1=23$ $p=12$ (c) $12p-5=25$ $p=25$

9. Solve: (a) $3n-2=46$ $n=16$ (b) $5m+7=17$ $m=2$ (c) $10p=100$ $p=10$ (d) $10p+10=100$ $p=9$
(e) $3s=-9$ $s=-3$ (f) $3s+12=0$ $s=-4$ (g) $2q-6=0$ $q=3$ (h) $2q+6=12$ $q=3$
(i) $\frac{20p}{3}=40$ (j) $\frac{3p}{10}=6$ $p=2$ (k) $\frac{3p}{4}=6$ $p=8$ (l) $\frac{-p}{3}=2$ $p=-6$

Worked examples

1 These solutions involve only one step.

a $m + 27 = 59$
$\quad -27 \quad -27$
$m = 59 - 27$
$\therefore m = 32$

b $a - 16 = 9$
$\quad +16 \quad +16$
$a = 9 + 16$
$\therefore a = 25$

c $3p = 23$
$\div 3 \quad \div 3$
$p = \frac{23}{3}$
$\therefore p = 7\frac{2}{3}$

d $\frac{x}{7} = 12$
$\times 7 \quad \times 7$
$x = 7 \times 12$
$\therefore x = 84$

2 The solutions of these equations involve two steps.

a $2a + 5 = 7$
$\quad -5 \quad -5$
$2a = 2$
$\div 2 \quad \div 2$
$\therefore a = 1$

b $\frac{x+3}{5} = 2$
$\times 5 \quad \times 5$
$x + 3 = 10$
$\quad -3 \quad -3$
$\therefore x = 7$

c $\frac{p}{3} - 6 = 15$
$\quad +6 \quad +6$
$\frac{p}{3} = 21$
$\times 3 \quad \times 3$
$\therefore p = 63$

d $10 - 3m = 25$
$\quad -10 \quad -20$
$-3m = 15$
$\div -3 \quad \div -3$
$\therefore m = -5$

Exercise 10:02

Foundation Worksheet 10:02
Solving equations 1, PAS3·1b
1 Solve these equations.
a $\boxed{x} + 7 = 10$ b $\boxed{y} - 2 = 6$
c $6 \times \boxed{y} = 18$ d $\boxed{y} \div 2 = 3$
2 Which of the numbers 2, 7 and 10 are solutions?
a $\boxed{y} + 4 = 11$ b $3 \times \boxed{y} = 30$

1 Solve each of these one-step equations.

a $x + 7 = 15 \to x = 8$
b $p + 9 = 11 \to p = 2$
c $y + 7 = 21 \to y = 14$
d $15 + n = 18 \to n = 3$
e $7 + q = 15 \to x = 8$
f $20 + k = 29 \to k = 9$
g $y - 5 = 3 \to y = 8$
h $m - 7 = 2 \to m = 9$
i $x - 10 = 3 \to x = 13$
j $5x = 15 \to x = 3$
k $7x = 42 \to x = 6$
l $3p = 36 \to p = 12$
m $12 = 3y \to y = 4$
n $18 = 9y \to y = 2$
o $24 = 8x \to x = 3$
p $\frac{n}{3} = 2 \to n = 6$
q $\frac{x}{5} = 3 \to x = 15$
r $\frac{m}{4} = 4 \to m = 16$

2 These one-step equations involve negative integers.

a $x + 3 = 2 \to x = -1$
b $m + 5 = 1$ $m = -4$
c $n + 6 = 4$ $n = -2$
d $7 + p = -1 \to p = -8$
e $6 + q = 0$ $q = -6$
f $10 + y = -2$ $y = -12$
g $a - 7 = -2 \to a = 5$
h $x - 10 = -3$ $x = 13$
i $p - 9 = -4$ $p = 5$
j $q - 2 = -1 \to q = 1$
k $y - 4 = -4$ $y = 8$
l $t - 6 = -2$ $t = 4$
m $3m = -6 \to m = -2$
n $8p = -16$ $p = -2$
o $6w = -6$ $w = -1$
p $-2n = 12 \to 6$
q $-3q = 27$ $q = -9$
r $-4m = -8$ $m = 2$
s $\frac{x}{3} = -2 \to x = -6$
t $\frac{m}{4} = -20$ $m = -80$
u $\frac{n}{7} = -28$ $n = -196$
v $x - (-2) = 5 \to x = 3$
w $-5 - y = 3$ $y = -8$
x $6 - (-x) = 3$ $x = 9$

> **Notice:**
> If $-a = 3$,
> then $a = -3$
> (Multiply both sides by -1)

3 The solution to an equation can be checked by substituting it into the equation. Check to see if the solution to each equation below is correct.

a $x + 16 = 43$ ✓
$x = 27$

b $9 + y = 13$ ✗
$y = 5$

c $m - 10 = 3$ ✓
$m = 13$

d $n - 12 = 15$ ✓
$n = 27$

e $6 - q = 4$ ✓
$q = 2$

f $8 - y = 10$ ✗
$y = 2$

g $7m = 10$ ✓
$m = \frac{10}{7}$

h $3a = 2$ ✗
$a = \frac{3}{2}$

i $-5y = -15$ $y = -3$ j $\frac{x}{3} = 6$ $x = 18$ k $\frac{m}{5} = 10$ $m = 2$ l $\frac{n}{6} = 3$ $n = \frac{1}{2}$

4 Solving these equations involves two steps. Clearly show each step in your working. (All the answers are integers.)

a $2x + 1 = 5$ b $3a + 2 = 8$ c $2m + 7 = 13$
d $3n - 2 = 7$ e $5k - 1 = 24$ f $6t - 4 = 8$
g $6 + 5a = 26$ h $10 + 3w = 13$ i $12 + 4q = 16$
j $10 - 3x = 1$ k $15 - 2m = 11$ l $20 - 5q = 0$
m $15 = 2x - 3$ n $7 = 5y - 3$ o $10 = 2 - 4a$
p $4 - x = -7$ q $-3 - x = -2$ r $2 = 4 - x$

5 Solving these equations also involves two steps. (The solutions are all positive integers.)

a $\frac{3x}{2} = 3$ b $\frac{2x}{5} = 4$ c $\frac{3m}{4} = 6$
d $\frac{x+1}{4} = 3$ e $\frac{a+2}{3} = 1$ f $\frac{n+7}{2} = 10$
g $\frac{m-4}{2} = 5$ h $\frac{n-1}{3} = 6$ i $\frac{a-2}{4} = 10$
j $\frac{y}{3} + 1 = 3$ k $\frac{x}{4} + 2 = 1$ l $\frac{m}{5} - 3 = 2$
m $5 + \frac{a}{2} = 10$ n $2 + \frac{x}{3} = 8$ o $5 + \frac{m}{2} = 7$

$\frac{3x}{2} = 6$
$\times 2 \quad \times 2$
$3x = 30$
$\div 3 \quad \div 3$
$x = 10$

Multiply each term on *both sides* by the denominator.

6 The solutions to these equations involve fractions.

a $2x + 1 = 4$ b $2a + 5 = 10$
c $3m + 4 = 5$ d $5n + 2 = 8$
e $2p - 3 = 2$ f $4q - 1 = 2$
g $5n - 5 = 4$ h $6y - 3 = 1$
i $5 + 2k = 12$ j $7 + 3x = 9$
k $8 + 3a = 10$ l $1 + 2a = 4$
m $2m + 6 = 3$ n $5p + 7 = 1$
o $3 - 2a = 6$ p $5 - 3a = 1$

$3a + 2 = 6$
$-2 \quad -2$
$3a = 4$
$\div 3 \quad \div 3$
$a = \frac{4}{3}$ or $1\frac{1}{3}$

7 Now try this set of equations, which are either one- or two-step types.

a $5x = 35$ b $m + 7 = 11$ c $2x + 1 = 3$ d $5 - n = 0$
e $3a + 1 = 10$ f $a - 2 = -3$ g $y + 3 = 1$ h $3m - 1 = 5$
i $-4x = 16$ j $5 + 3n = 10$ k $3x = 2$ l $6 - q = 10$
m $\frac{x}{5} = 10$ n $\frac{3p}{2} = 6$ o $\frac{x+2}{5} = 1$ p $\frac{a}{3} - 4 = 2$
q $6 + \frac{a}{3} = 8$ r $\frac{5+a}{2} = 1$ s $\frac{m-4}{7} = 6$ t $\frac{5m}{2} = 3$
u $6 - 2a = 8$ v $3p + 5 = 2$ w $\frac{q+6}{2} = 1$ x $\frac{3-x}{2} = 5$

Fun Spot 10:02

Work out the answer
answer.

Solve these equations.

T $x + 8 = 15$
T $3 + x = 1$
A $x + x = 1$
E $8a = 0$
U $60 - n = 0$
T $n = 20 - n$
O $6 + x = 2$
I $-3 - 3 = a$
B $-a = -8$
T $10 - n = -2$

Examples: Factorise the following.

1. $2a + 6 = 2(a+3)$

2. $a^2 + 2a = a^2+2a$

3. $8x^2 - 6x$ Unable to read - poor quality

Note:

4. $2a^3b^2 + 4ab^2$

5. $16a^4bc - 12a^2b^2$

B) Factorising out the negative

When dealing with negative terms, you must think about the following rules:

1. If the first term is negative, factorise out the (-) sign as well.
2. When taking out the (-) sign, it flips all the signs of the terms.

E.g. $-x - 3 = -(x + 3)$

Examples: Factorise the following.

1. $-5a - 26$

2. $-4x - 8$

3. $-4x^2 - 2x$

4. $-5m^2p + mp^3$

C) Complicated Factorisation

Whenever two terms have a common bracket, the entire bracket can be taken out.

You must ask:

1. "What's common"
2. "What's left"

Unable to read - poor quality

Examples: Factorise the following.

1. $4(3x - 2) + x(3x - 2)$ Already factorized

1. $4(3x-2) + x(3x-2) = 12x - 8 + 3x^2 - 2x =$

Note:

2. $2(x + 1) - x(x + 1)$

Note:

3. $x(x^2 + 1) + (x^2 + 1)$

Note:

4. $2x(x^2 - 1) - (x^2 - 1)$

D) Simplifying Fractions with Brackets

Simplifying fractions with brackets

$$\frac{2(3)}{2} = 3$$

$$\frac{2 + 3}{3} \neq 3$$

$$\frac{2(x + 1)}{2} = x + 1$$

$$\frac{2 + (x + 1)}{2} \neq x + 1$$

Quality too low

Classwork: Simplify the following.

1. $\frac{2x}{5} - \frac{x}{3} = \frac{6x}{15} - \frac{5x}{15} = \frac{x}{15}$

2. $\frac{1}{3}y - \frac{5}{12}y = \frac{4}{12}y - \frac{5}{12}y = \frac{-1}{12}y$

3. $\frac{4}{5}ab - \frac{1}{3}ab = \frac{12}{15}ab - \frac{5}{15}ab = \frac{7}{15}$

4. $\frac{x+3}{4} + \frac{2x-1}{3} = \frac{3x+9}{12} + \frac{8x-4}{12} = \frac{11x+5}{12}$

5. $\frac{6x+1}{12} + \frac{2x+3}{8} = \frac{12x+2}{24} + \frac{6x+9}{24} = \frac{18x+11}{24}$

6. $\frac{2x+1}{2} - \frac{x+1}{3} = \frac{6x+3}{6} - \frac{2x+2}{6} = \frac{4x+1}{6}$

7. $\frac{3x-1}{9} - \frac{4-x}{6} = \frac{6x-2}{18} - \frac{12-3x}{18} = \frac{9x-14}{18}$

8. $\frac{3}{2y} - \frac{4}{y} = \frac{3}{2y} - \frac{8}{2y} = \frac{-5}{2y}$

9. $\frac{b}{a^3} - \frac{2}{a^2} = \frac{b-2a}{a^3}$

10. $\frac{3}{x^3y^2} - \frac{2}{x^2y^3} = \frac{3y}{x^3y^3} - \frac{2x}{x^3y^3} = \frac{3y-2x}{x^3y^3}$

11. $\frac{4}{5x^2y^2} - \frac{3}{10xy^3} =$

12. $\frac{4}{15a^2bc^3} - \frac{1}{6ab^2c^3}$

How do we do this

Practice

1 Expand.

a $3(2x - 1) = 6x - 3$

b $-2(5pq + 4q^2) =$ $\rightarrow -10pq - 4q^2$

c $-(3xy - 2y^2) = -3xy + 2y^2$

2 Expand and simplify.

a $7(3x + 5) + 6(2x - 8)$

b $8(5p - 2) - 3(4p + 9)$

c $9(3s + 1) - 5(6s - 10)$

d $2(4x - 3) - (3x + 5) =$ $\rightarrow 8x$

Watch out!

When multiplying (or dividing) positive and negative numbers, if the signs are the same the answer is '+'; if the signs are different the answer is '–'.

3 Expand.

a $3x(4x + 8) = 12x^2 + 8x$

b $4k(5k^2 - 12)$

c $-2h(6h^2 + 11h - 5) =$ $\rightarrow -12h^3 - 22h^2 + 10h$

d $-3s(4s^2 - 7s + 2)$

4 Expand and simplify.

a $3(y^2 - 8) - 4(y^2 - 5)$

b $2x(x + 5) + 3x(x - 7)$

c $4p(2p - 1) - 3p(5p - 2)$

d $3b(4b - 3) - b(6b - 9)$

5 Expand $\frac{1}{2}(2y - 8)$

6 Expand and simplify.

a $13 - 2(m + 7)$

b $5p(p^2 + 6p) - 9p(2p - 3)$

7 The diagram shows a rectangle.

Write down an expression, in terms of x, for the area of the rectangle.

Show that the area of the rectangle can be written as $21x^2 - 35x$

8 Expand and simplify.

a $(x + 4)(x + 5)$

b $(x + 7)(x + 3)$

c $(x + 7)(x - 2)$

d $(x + 5)(x - 5)$

e $(2x + 3)(x - 1)$

f $(3x - 2)(2x + 1)$

g $(5x - 3)(2x - 5)$

h $(3x - 2)(7 + 4x)$

i $(3x + 4y)(5y + 6x)$

j $(x + 5)^2$

k $(2x - 7)^2$

l $(4x - 3y)^2$

Extend

9 Expand and simplify $(x + 3)^2 + (x - 4)^2$

10 Expand and simplify.

a $\left(x + \frac{1}{x}\right)\left(x - \frac{2}{x}\right)$

b $\left(x + \frac{1}{x}\right)^2$

Sequences from patterns

5_
11

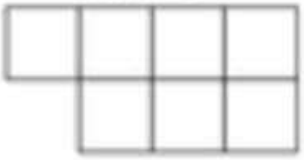
1. Here is a pattern of squares

1 : 3
2 : 5
3 : 7

Pattern 1

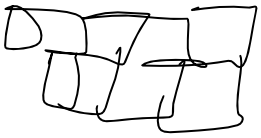
Pattern 2

Pattern 3



a) Draw pattern 4

Pattern 4:

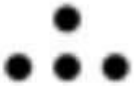


b) How many squares will there be in pattern number 13? 27 squares

c) Which pattern has 27 squares? 13

2. Here is a pattern of dots

Pattern 1



Pattern 2

Pattern 3

a) Draw pattern 4

b) How many dots will there be in pattern 15?

c) Jack says there are 70 dots in pattern 20. Is he right? You must explain your answer.

3. Here is a made from bricks

Pattern 1

Pattern 2

Pattern 3

a) Draw pattern 4

b) How many bricks will there be in pattern number 8?

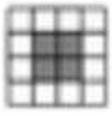
c) Which pattern will have 49 bricks?

d) How many bricks will there be in pattern *n*?

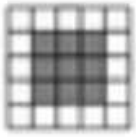
4. Here is a made from white and grey tiles.

Pattern 1

Pattern 2



Pattern 3



a) Draw pattern 4

b) How many white tiles will there be in pattern number 10?

c) How many grey tiles will there be in pattern 8?

d) Miles has 49 grey tiles and 32 white tiles. Which pattern can he make?

e) Emma has 90 grey tiles and 42 white tiles. She says she can make pattern number 10. She is incorrect, explain why.

f) How many more grey and white tiles will Emma need to make pattern number 10.

First Term & Common Difference of an Arithmetic Sequence

Write the first term(a) and the common difference(d) of each arithmetic sequence.

1) $-700, -750, -800, -850, -900, \ldots$

a = _____ ; d = _____

2) $9.1, 2.6, -3.9, -10.4, -16.9, \ldots$

a = _____ ; d = _____

3) $27.4, 23.4, 19.4, 15.4, 11.4, \ldots$

a = _____ ; d = _____

4) $1, \frac{3}{4}, \frac{1}{2}, \frac{1}{4}, 0, \ldots$

a = _____ d = _____

5) $\frac{2}{5}, \frac{16}{15}, \frac{26}{15}, \frac{12}{5}, \frac{46}{15}, \ldots$

a = _____ ; d = _____

6) $2, 16, 26, 30, \ldots$

a = _____ ; d = _____

7) $18, 9, 0, -9, -18, \ldots$

a = _____ ; d = _____

8) $-\frac{7}{3}, -\frac{16}{3}, -\frac{25}{3}, -\frac{28}{3}, -\frac{37}{3}, \ldots$

a = _____ ; d = _____

9) $\sqrt{2}, -\sqrt{18}, -\sqrt{98}, -\sqrt{242}, -\sqrt{450}, \ldots$

a = _____ ; d = _____

10) $-33, -48, -63, -78, -93, \ldots$

a = _____ ; d = _____